

On the Effects of Sea Surface Skewness Function and Its Parameter Related to Small-Scale Waves on Radar Backscattering: A Numerical Study

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Abstract—Sea surface skewness function used to describe the horizontal asymmetry of the sea surface is related to the third-order cumulant of rough sea surface, which is a key input for the modeling of radar scattering from non-Gaussian sea surface. The horizontal asymmetry of the sea surface is caused by small-scale sea waves and affects the radar echoes when the radar views the sea surface along different wind directions. However, currently, the sea surface skewness function and its parameters related to small-scale waves are diversified and there is no clear guidance in their choice. To this end, this letter first evaluated the effects of different skewness functions and small-scale wave components on the radar backscattering from non-Gaussian sea surface based on the first-order small slope approximation (SSA1) model. Then, the optimal skewness function and its parameter related to small-scale waves were determined by comparing the simulated backscattering contributed by different skewness functions with various small-scale wave components with radar measurements. Further, the performance of the SSA1 model with the optimal skewness function was compared with that of the two-scale model (TSM) including hydrodynamic modulation. Clarifying the effect patterns of skewness function and small-scale wave components on the radar scattering from non-Gaussian seas is instructive for us to model the radar scattering from the real sea surface, serving the optimization of radar observation scheme for the effective retrieval of sea surface wind field.

Index Terms—First-order small slope approximation (SSA1), non-Gaussian sea surface, radar scattering, skewness function.

I. INTRODUCTION

THE ocean surfaces are equipped with multiscale roughness generated by long waves on which ride many short waves and capillary waves driven by local wind field [1], [2]. The sea surface height deviating from the mean sea level is usually a non-Gaussian distribution and thus its higher-order (greater than second-order) cumulants are not equal to zero. The sea wave power spectrum belongs to the second-order statistic because it is the Fourier transform of the autocorrelation function of sea surface height, and thus the electromagnetic (EM) scattering simulation based on the sea wave spectrum can only reflect the geometric information of Gaussian sea surface [3].

Received 1 August 2024; revised 5 September 2024; accepted 15 September 2024. Date of publication 18 September 2024; date of current version 1 October 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 42006158 and Grant 42271420 and in part by the Natural Science Foundation of Jiangsu Province under Grant BK20200698. (Corresponding author: Jinghai Xu.)

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Digital Object Identifier 10.1109/LGRS.2024.3463207

To obtain the information of non-Gaussian sea surface from radar echoes, it needs to add the higher order statistics in the modeling of sea surface radar scattering. The bicorrelation function is related to the third-order cumulant of the non-Gaussian sea surface and its Fourier transform is referred to as bispectrum [4], [5]. As far as I know, Chen et al. [4] first introduced the bicorrelation function into the scattering model to explain the upwind-downwind asymmetry of radar scattering cross section. The bicorrelation function is a real function and can be decomposed into the sum of an even function and an odd function. The even function is called the symmetric part of the bicorrelation function, describing the vertical asymmetry of the sea surface; while the odd function is called the asymmetric part, describing the horizontal asymmetry of the sea surface [4]. Of the two parts, only the asymmetric part is essential to simulate the upwind-downwind asymmetry and is called the “skewness function”; while the symmetric part has a similar role, but very small on an order of magnitude, to that of the sea surface autocorrelation function in the modeling of sea surface radar scattering [6]. Given the small level contributed by the vertical asymmetry of the sea surface to radar scattering and the responsibility of horizontal asymmetry to the difference of radar scattering at upwind and downwind directions, in this letter, we only focus on the effect of the asymmetric part of bicorrelation function, i.e., skewness function, on radar scattering from non-Gaussian sea surface.

The sea surface skewness is due to the small-scale capillary waves, which have been proved by the measurements of oil-covered sea surface [6], [7]. This is because the oil slick tends to suppress the capillary waves and reduces the skewness, which means the skewness is related to the small-scale waves. So, it naturally poses a question that how to divide the sea surface into large- and small-scale waves for the radar scattering simulation of the sea surface. This division is not uniform yet, varying according to the respective research objectives (e.g., [8], [9]). Although previous researchers have studied the modeling of sea surface scattering from non-Gaussian sea surfaces, they either mainly focus on the closed-form expression of the skewness function [10], or discuss the skewness functions with the fixed small-scale waves [6] or the small-scale waves for given skewness function [11]. Therefore, to better use the skewness function, it is necessary to comprehensively explore the effects of the skewness function and small-scale sea waves on the radar scattering to guide us in choosing the appropriate sea surface skewness and its parameter in the modeling of radar scattering from the non-Gaussian sea surface. To achieve this objective, the widely used first-order small slope approximation (SSA1)

model including the skewness function [6] is adopted to simulate the radar scattering from non-Gaussian sea surface.

The following is the organization of this letter. The SSA1 model along with the sea wave spectrum and skewness function are reviewed briefly in the section of methodology. Next, the normalized radar cross sections (NRCSs) contributed by different skewness functions under different small-scale wave components were simulated and analyzed to clarify their effect patterns on radar scattering. On this basis, the optimal skewness function and sea wave components were determined by comparing these simulated NRCSs with radar measurements. Finally, the comparison between the performance of SSA1 with the optimal skewness function and that of the two-scale model (TSM) including hydrodynamic modulation was conducted.

II. METHODOLOGY

A. Elfouhaily Sea Spectrum Model

The 2-D directional sea spectrum is expressed as the product of the omnidirectional spectrum $S(K)$ by the angular spreading function (ASF) $\Phi(K, \phi)$ [2] and defined in polar coordinates as

$$S(K, \phi) = \frac{1}{K} \cdot S(K) \cdot \Phi(K, \phi) \quad (1)$$

where ϕ is the azimuth angle of the wavenumber relative to the upwind direction. The variance and 2-D autocorrelation function of the sea surface are related to the omnidirectional spectrum and 2-D sea spectrum, respectively, and are defined as follows:

$$\sigma_z^2 = \int_0^\infty S(K) dK \quad (2)$$

$$C(r, \varphi) = \int_0^\infty \int_0^{2\pi} S(K, \phi) \cdot \exp(jKr \cos(\varphi - \phi)) K dK d\phi \quad (3)$$

where φ is the azimuth angle of the 2-D correlation function relative to the upwind direction.

B. Sea Surface Skewness Function

With the help of a stereo photographic technique [12], the skewness function can be modeled in the following form:

$$S_a(r, \varphi) = -\sigma_{zs}^3 \cdot r^3 / s^3 \cdot \exp[-(r/s)^p] \cos(\varphi) \quad (4)$$

where 1) σ_{zs} is the root mean square (rms) height of small-scale sea waves and is calculated by using $\sigma_{zs}^2 = \int_{c \times k}^\infty S(K) dK$ where k is the wavenumber of radar frequency and c is the coefficient of dividing wavenumber; 2) p represents the *power-index* of the skewness function, e.g., $p = 1$ in [5] and [12], whereas in [5], $p = 2$; and 3) s is called “skewness length” analogous to the correlation length in the autocorrelation function and has the following expression [6]:

$$s = (6/c_{03})^{1/3} (\sigma_{zs}/\sigma_{sx}) \quad (5)$$

where σ_{sx} is the surface rms slope in the upwind direction and c_{03} is the coefficient in the slope probability density function. They are related to the wind speed at 12.5 m above sea level $U_{12.5}$, i.e., $c_{03} = (3.3 \times U_{12.5} - 4) \times 10^{-2}$, $\sigma_{sx}^2 = (3.16 \times U_{12.5}) \times 10^{-3}$ [6]. Assuming a neutral stability atmosphere, the relationship between $U_{12.5}$ and U_{10} is

$$U_h = U_{10} \ln(12.5/z_0) / \ln(10/z_0) \text{ with } z_0 = 0.000684/u^* + 0.00428(u^*)^2 - 0.000443 \text{ [13] and the friction velocity } u^* = U_{10}(10^{-3} \times (0.81 + 0.065U_{10}))^{1/2}. \text{ [14].}$$

Thus, there is only one parameter, i.e., rms height of small-scale sea waves or the c -value, in the skewness function. For more details about the derivation of the skewness function, readers can refer to [6].

C. SSA1 Model With Sea Surface Skewness Function

Considering a plane wave with an EM wavenumber of k impinges on a rough sea surface at an incidence angle and azimuth angle of (θ_i, ϕ_i) , the scattered wave is with the scattering angles of (θ_s, ϕ_s) . The radar-measured scattering amplitude is associated with sea surface roughness illuminated by the antenna, the incidence and scattering angles, and their corresponding azimuth angles. The mathematical expression for NRCSs calculated by the SSA1 model with and without the skewness function are given, respectively, as follows [6]:

$$\sigma_{0pq}^{\text{SSA1s}}(\mathbf{k}_s, \mathbf{k}_i) = \frac{1}{\pi} \left| \frac{2q_i q_s B_{pq}(\mathbf{k}_s, \mathbf{k}_i)}{Q} \right|^2 \cdot \exp(-\sigma_z^2 Q^2) \cdot \int \{ \exp[Q^2 C(\mathbf{r}) - jQ^3 S_a(\mathbf{r})] - 1 \} \cdot \exp[-j(\mathbf{k}_s - \mathbf{k}_i) \cdot \mathbf{r}] d\mathbf{r} \quad (6)$$

$$\sigma_{0pq}^{\text{SSA1}}(\mathbf{k}_s, \mathbf{k}_i) = \frac{1}{\pi} \left| \frac{2q_i q_s B_{pq}(\mathbf{k}_s, \mathbf{k}_i)}{Q} \right|^2 \cdot \exp(-\sigma_z^2 Q^2) \cdot \int \{ \exp[Q^2 C(\mathbf{r})] - 1 \} \cdot \exp[-j(\mathbf{k}_s - \mathbf{k}_i) \cdot \mathbf{r}] d\mathbf{r}. \quad (7)$$

In (6) and (7), $\sigma_{0pq}^{\text{SSA1s}}$ indicates the scattering from the sea surface contains the contribution from the skewness function, whereas $\sigma_{0pq}^{\text{SSA1}}$ does not; the subscripts p and q denote the received and transmitted polarization states, respectively; $\{\mathbf{k}_i, \mathbf{k}_s\}$ are the horizontal projections of the incident and scattered wave vectors and $\{q_i, q_s\}$ are the absolute values of their vertical projections. $Q = q_i + q_s$, and $B_{pq}(\mathbf{k}_s, \mathbf{k}_i)$ is the polarization-related matrix. σ_z is rms height of sea surface expressed in (2). $C(\mathbf{r})$ and $S_a(\mathbf{r})$ are the 2-D autocorrelation function and skewness function of the sea surface expressed in (3) and (4), respectively.

III. NUMERICAL RESULTS AND DISCUSSION

In this section, the NRCS is simulated under the assumption that the sea is fully developed. The Elfouhaily spectrum [2] describing the Gaussian information of sea surface and the skewness function describing the asymmetry of the non-Gaussian sea surface with the mathematical expression (4) are adopted to simulate the NRCSs to be needed. The permittivity of sea water is calculated by the Debye equation [15] with fixed seawater temperature and salinity at 20 °C and 35‰, respectively. In addition, it should be noted that the SSA1 model does not well consider the effects of sea surface curvature [16] and steep breaking waves [17] resulting in an underestimation at HH-polarization in backscatter, while VV-polarization is almost unaffected. Therefore, the modeled VV-polarization is used to carry out the analysis of the effects of skewness function and small-scale sea wave components on radar backscattering.

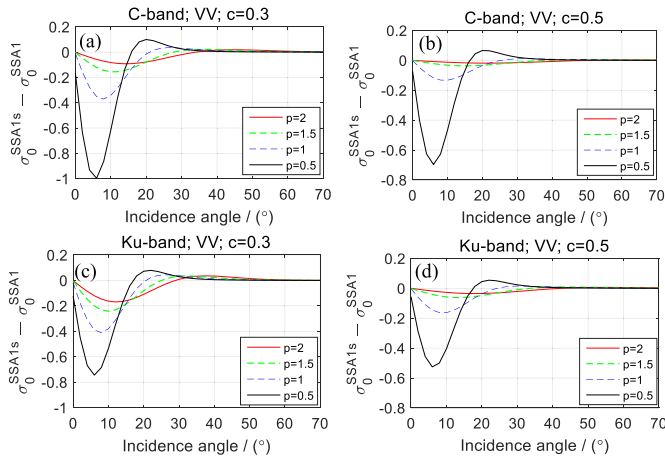


Fig. 1. Contributions from skewness function with different p - and c -values to radar backscattering in terms of radar incidence angles at the upwind direction and wind speed of 10 m/s for (a) and (b) C- and (c) and (d) Ku-bands.

A. Effect Pattern of Skewness Function and Small-Scale Wave Components on Radar Backscattering

Fig. 1 shows the contributions of skewness function with different power-index [i.e., p -value in (4)] and different small-scale wave components determined by c -value to radar backscattering in terms of radar incidence angles at C-(5.3 GHz) and Ku-(14 GHz) bands when wind speed is 10 m/s and the radar look direction is in line with the upwind direction ($\phi = 0^\circ$).

It can be seen from Fig. 1 that the contribution from the skewness function to backscattering at a small incidence angle is negative and the range of incidence angle corresponding to the negative contribution will gradually increase with the increase of p -value. Besides, the levels of negative contribution at small incidence angles are higher than that of positive contribution at medium incidence angles. At incidence angles greater than 40° , the difference between the contributions from these skewness functions to radar backscattering becomes small gradually. As the c -value increases, the level of contribution from the skewness function to radar backscattering diminishes, and simultaneously the range of incidence angle corresponding to the negative contribution from the skewness function with a larger p -value enlarges [Fig. 1(a) and (b)]. When the p -value is 0.5, the range of incidence angle corresponding to the negative contribution is insensitive to the c -value. At the Ku-band, the same effect pattern of skewness function and small-scale waves on radar scattering as that at the C-band is observed [Fig. 1(c) and (d)].

Since the contributions from the skewness function and sea spectrum to radar backscattering are both larger at small incidence angles, the contribution from the skewness function relative to that from the sea spectrum is essential to weigh the influence of skewness function on radar scattering at different incidence angles. Hence, we take the skewness function with a p -value equal to 2 as an example to simulate the relative contributions at the upwind direction for different small-scale wave components. The relevant simulations at wind speeds of 5 and 15 m/s for C- and Ku-bands are shown in Fig. 2.

It can be seen in Fig. 2 that the relative contribution from the skewness function at a small incidence angle is small for a large c -value. So, the contribution from the skewness function with a large c -value at small incidence angles can be treated as noise. The larger relative contributions appear at the medium

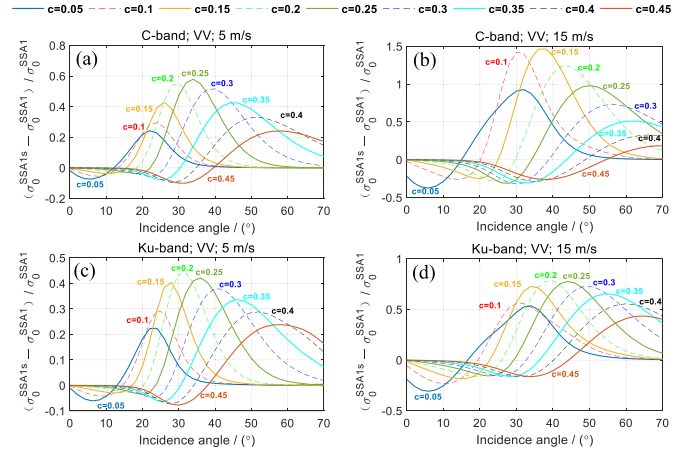


Fig. 2. Contributions from skewness function with $p = 2$ to radar scattering relative to that from sea spectrum at the upwind direction in terms of radar incidence angles for different c -values at wind speeds of 5 and 15 m/s for (a) and (b) C- and (c) and (d) Ku-bands.

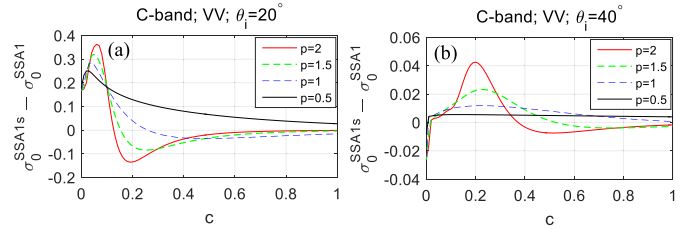


Fig. 3. Contributions from skewness function with different p -values to radar backscattering varying with c -values at upwind direction and wind speed of 10 m/s for C-band with radar incidence angles of (a) 20° and (b) 40° .

incidence angles and the range of the medium incidence angles will shift toward the direction of incidence angle increase with the increase of c -value or wind speed. The same effect pattern of skewness function on radar scattering can be also observed at the Ku-band. For a given c -value, as the radar frequency increases, the range of the radar incidence angle corresponding to a larger relative contribution moves to the direction of the incidence angle decrease. Besides, it is worth noting that a specific scattering contribution from the skewness function may correspond to two different c -values. It indicates that parameter c in the skewness function cannot be determined solely by the radar measurement at one incidence angle. To show this more clearly, we take C-band as an example to simulate the scattering contribution from different skewness functions varying with c -values at upwind direction and wind speed of 10 m/s for incidence angles of 20° and 40° , which are shown in Fig. 3.

In Fig. 3, one can see that for a fixed radar incidence angle, there may be two different c -values responsible for a definite scattering contribution. Besides, as the p -value decreases, the contribution from sea surface skewness to radar backscattering becomes less and its sensitivity to the c -value also decreases. For the skewness function with a p -value of 0.5, its contributions to radar scattering are almost no change at an incidence angle of 40° .

B. Comparisons Between Simulated NRCS With GMF

Based on the analysis of the influence of skewness function and small-scale waves on radar backscattering in Section III-A, the skewness function and small-scale wave components responsible for radar measurements can be

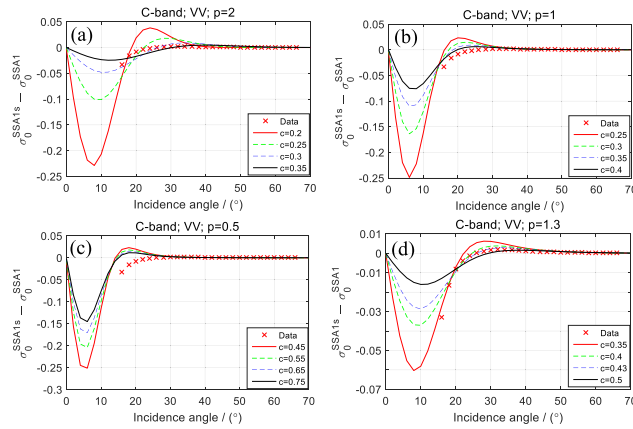


Fig. 4. Comparisons between radar measurements and the simulated NRCSs at C-band and wind speed of 5 m/s along upwind direction in terms of incidence angle based on different c -values and the skewness function with (a) $p = 2$, (b) $p = 1$, (c) $p = 0.5$, and (d) $p = 1.3$.

determined by comparing the modeled NRCSs with the corresponding radar measurements. The radar measurements for C- and Ku-bands are derived from the Geophysical Model Function (GMF) model CMOD7 [18] and NSCAT-4DS [19], respectively. The contribution from the skewness function to radar backscattering is equal to half of the difference between the radar-measured NRCSs at upwind and downwind directions (i.e., $(GMF_{upwind} - GMF_{downwind})/2$) [6]. Next, we take the NRCSs simulated at C-band and wind speed of 5 m/s as an example to demonstrate how to determine the optimal skewness function and small-scale wave components. The comparisons between simulations and reference data are shown in Fig. 4.

First, the contributions from the common skewness functions (i.e., Gaussian-type, $p = 2$; exponential-type, $p = 1$) to radar backscattering are analyzed. According to the analysis for Fig. 1, one knows the NRCSs simulated based on Gaussian-type skewness function are sensitive to c -value. As can be seen from Fig. 4(a), a small change in c -value for $p = 2$ will lead to a large difference in simulation results especially at incidence angles between 20° and 40° , and the simulations based on different c -values agree well with radar measurements only at partial incidence angles. Similarly, for the exponential-type skewness function [Fig. 4(b)], one can see the sensitivity of simulated results to the c -value decreases, which means the simulated data cannot well match the reference data overall no matter how the c -value is adjusted. When the p -value decreases further to 0.5 [Fig. 4(c)], the insensitivity of simulation results to the c -value becomes more obvious. Given these, we adjust p - and c -values to make the simulated data match the reference data with a relatively small root-mean-square error (RMSE), especially for the simulations at incidence angles greater than 20° . Because the scattering contributions from the skewness function at incidence angles less than 20° are not important relative to that from the sea spectrum [see Fig. (2)]. Finally, we find that the combination of ($p = 1.3$, $c = 0.43$) makes the simulations agree with reference data well overall [Fig. 4(d)]. In a similar way, the optimal combinations of p - and c -values which make the match between simulations and reference data have a small RMSE are determined at C- and Ku-bands for the wind speeds of 5, 10, and 15 m/s. The match results are shown in Fig. 5.

It can be seen from Fig. 5 that these optimal combinations of p - and c -values can make the NRCS simulations match

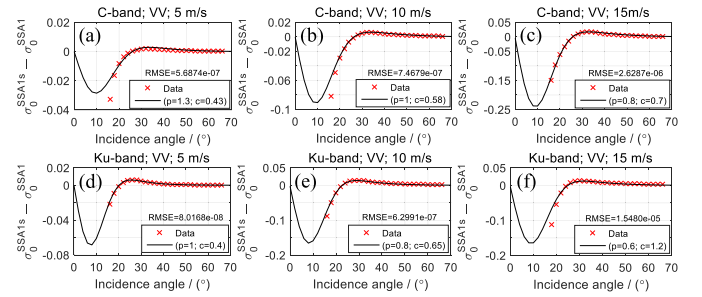


Fig. 5. Optimal combinations of p - and c -values which make the simulated contribution from skewness function match the radar-measured data well overall at wind speeds of 5, 10, and 15 m/s for (a)–(c) C- and (d)–(f) Ku-bands.

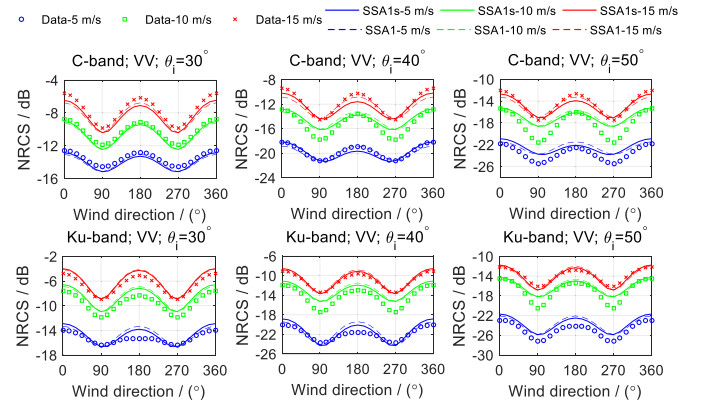


Fig. 6. Comparisons between radar measurements and the simulated NRCSs with and without the optimal skewness function in terms of wind direction at wind speeds of 5, 10, and 15 m/s for C- and Ku-bands when radar incidence angles are 30° , 40° , and 50° , respectively.

the reference data well except at these noncritical incidence angles less than 20° . To examine the contribution of skewness function to radar scattering at various wind directions, we also compared radar measurements with the simulated NRCSs with and without the optimal skewness function shown in Fig. 5 in terms of wind direction at wind speeds of 5, 10, and 15 m/s for C- and Ku-bands when incidence angles are 30° , 40° , and 50° , respectively, which are shown in Fig. 6.

It can be observed from Fig. 6 that these optimal skewness functions can be used to capture the difference of radar backscattering at different wind directions especially in upwind and downwind directions while it does not work for the sea wave spectrum. At a wind speed of 5 m/s, the deviations between radar measurements and the simulated data at crosswind direction for incidence angles of 40° and 50° may result from the sea wave spectrum, because the skewness function has no contribution to the NRCSs at crosswind direction. Although the contribution from skewness function to radar backscattering is smaller than that from the sea spectrum, it is helpful to distinguish wind directions, especially for upwind and downwind directions. The optimal skewness function along with its parameter related to small-scale waves shown in Fig. 5 are summarized and listed in Table I.

C. Comparisons Between SSA1s and TSM With Hydrodynamic Modulation (TSM + HM)

In addition to the skewness function, the hydrodynamic modulation combining the TSM is another theory to predict the asymmetry of NRCSs measured in upwind and downwind directions. To compare with TSM + HM, two sets of simulated data based on TSM + HM and Elfouhaily spectrum at C-band

TABLE I

SKEWNESS FUNCTION AND ITS PARAMETERS RESPONSIBLE FOR RADAR MEASUREMENTS AT C- AND KU-BANDS FOR WIND SPEEDS OF 5-, 10-, AND 15-M/S

Wind speed (m/s)	C-band ($k=111$ Rad/m)			Ku-band ($k=293$ Rad/m)		
	p	c	$c \times k$	p	c	$c \times k$
5	1.3	0.43	48	1	0.4	117
10	1	0.58	64	0.8	0.65	190
15	0.8	0.7	78	0.6	1.2	352

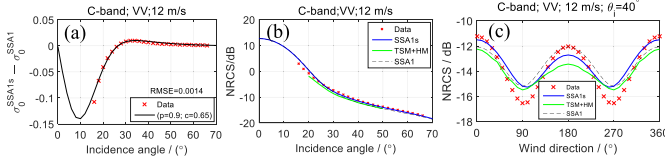


Fig. 7. (a) Optimal skewness function at C-band and wind speed of 12 m/s determined by comparing with COMD7. (b) Comparisons between the simulated NRCSs based on the SSA1s, SSA1, TSM + HM, and COMD7 varying with incidence angles at wind speed of 12 m/s. (c) Similar to (b) but varying with wind directions at an incidence angle of 40°.

and wind speed of 12 m/s were extracted from the existing literature [20]: one that varies with incidence angles [20, Fig. 4(a)] and the other that varies with wind directions [20, Fig. 7(a)]. Before conducting the comparison, it needs to determine the optimal skewness function for the C-band at a wind speed of 12 m/s. The same strategy mentioned in Section III-B is adopted to determine the optimal skewness function by comparing it with COMD7 and the comparison result is shown in Fig. 7(a). The optimal skewness function with the combination of ($p = 0.9$, $c = 0.65$) is finally determined. Based on the optimal skewness function, the NRCSs in terms of incidence angles and wind directions are simulated by using the SSA1s expressed in (6). The comparisons between them and those calculated based on SSA1, TSM + HM, and COMD7 are shown in Fig. 7(b) and (c), respectively.

One can see in Fig. 7(b) that the simulated NRCSs based on SSA1 with the optimal skewness function are more consistent with COMD7 at incidence angles greater than 30°. From Fig. 7(c), it can be seen that SSA1s and TSM + HM can both predict the asymmetry of NRCSs observed at upwind and downwind directions while the SSA1 cannot do; what is more, it also indicates the performance of SSA1 with the optimal skewness function is better than that of TSM with the hydrodynamic modulation shown in [20] in the accuracy of prediction overall.

IV. CONCLUSION

This letter investigates the effect pattern of the skewness function and its parameter related to small-scale wave components on radar scattering from non-Gaussian sea surfaces. Based on the effect pattern, the optimal skewness function and its parameters are determined by comparing the simulated NRCSs with radar measurements at different radar bands and wind speeds. The simulation results show that: 1) no matter how c -value is adjusted, the NRCSs simulated based on the commonly used Gaussian- and exponential-skewness functions may not match the radar measurements well at a wide range of incidence angles for some given radar bands and wind

speeds; 2) as radar frequency and wind speed increase, the p -value of the optimal skewness function decreases and its corresponding small-scale wave components also shift toward the direction of the increase of wavenumber; 3) the determined optimal skewness function is helpful in the simulation of radar scattering from non-Gaussian sea surface for the optimization study of radar observation scheme for the inversion of sea surface wind field.

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